



NanoPhotometer[®] N120/NP80/N60/N50/C40

User Manual

Version 4.6.7

Software Version 4.6m



4. NANOPHOTOMETER® APPLICATIONS

The NanoPhotometer® comes with pre-programmed applications as well as the ability to create custom applications. Any application method can be selected by tapping the icon once or pressing the icon (computer based software).

Note: Within each method selection of NanoVolume or cuvette mode (NP80/C40) is possible. The “Change to Cuvette” or “Change to NanoVolume” button is located at the bottom of the parameter area.



NUCLEIC ACIDS

METHOD OVERVIEW

Nucleic acids in solution absorb light with a peak in the ultraviolet region of 260 nm. For determination of nucleic acid concentration in solution the absorbance at wavelength 260 nm is used along with the Beer-Lambert law. In addition to calculating concentrations of nucleic acids, absorbance measurements are also useful for estimating purity of nucleic acids by calculating the 260/280 nm and 260/230 nm ratios. Further, it is possible to determine the degree of labeling of nucleic acids with probes including fluorescent dyes.

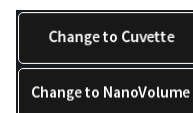
Sample Control™ gives useful information about sample conditions. It recognizes air bubbles, sample impurities, turbidity, lint residues and potential contaminations. If Sample Control™ detects any inconstancy an alert icon  is shown in the result/table area. A push on the alert icon shows additional information about the inconstancy.

MEASUREMENT PROTOCOL SINGLE SAMPLE

1. Select the Nucleic Acids icon on the home screen.



2. To change between NanoVolume and cuvette application (NP80/C40 only), press the Change to Cuvette/NanoVolume button below the parameter area.



3. To change the nucleic acid type push on dsDNA and a list with available options opens on the right side.

Options are: dsDNA, ssDNA, RNA miRNA, miRNA Sequence, Oligo, Oligo Sequence and Custom (see Table 1 on page 66).

- The miRNA sequence and Oligo sequence options allow the sequence to be entered and the extinction coefficient will be automatically calculated. A minimum of 4 bases is required.
- The default setting for Custom is 50. The extinction coefficient ($1/\epsilon$) can be set in a range of 15 - 150. It is possible to enter a name for the custom nucleic acid factor for documentation.
- The default setting for Custom Molar Extinction Coefficient (Mol. Ext. Coeff.) is 10 and be set within a range of 0 - 15. It is possible to enter a name for the custom molar extinction coefficient for documentation.

dsDNA	50	dsDNA	50	ssDNA	37
Volume	1 µl - 2 µl	RNA	40	miRNA	33
Units	ng/µl	miRNA Seq.	Please enter miRNA sequence		
Background Correction		Oligo	33		
Air Bubble Recognition	☐	Oligo Seq. DNA	Please enter DNA Oligo sequence		
Dye Label		Oligo Seq. RNA	Please enter RNA Oligo sequence		
Manual Dilution		Custom	Please enter Name	Please enter Factor	
		Custom (Mol. Ext.Coeff.)	Please enter Name	Please enter Mol. Ext. Coeff.(nM·1*cm-1)	

4. For NanoVolume application:

Select the volume of sample to be applied.

dsDNA	50	1-2 µl sample volume	Autoranging
Volume	1 µl - 2 µl	0.3 µl sample volume	Dilution 140 / 0.07 mm path

Note: 1 - 2 µl (default): automatic path length setting; 0.3 µl measures only the 0.07 mm path length (for samples with concentrations > 420 ng/µl dsDNA)

For cuvette application:

Select the path length depending on the used cuvette.

Options are: 0.5 mm, 1 mm, 2 mm, 5 mm and 10 mm

dsDNA	50	Pathlength	0.5 mm
Pathlength	10 mm	Pathlength	1 mm
Heat To 37°C	☐	Pathlength	2 mm
Units	ng/µl	Pathlength	5 mm
		Pathlength	10 mm

If it is desired to heat the sample to 37°C use the toggle switch to turn on cell holder heater. When the cuvette holder has reached 37°C the toggle color changes to green.

Note: Only available for cuvette applications (NP80 and C40).

For NanoVolume application with submicroliter cell (C40 only):

Select a Lid depending on your sample concentration for concentration ranges of different lids (see page 52 Submicroliter Cell Measurements Basics).

Required sample volume for each lid is shown.

dsDNA	50	Lid 5	3.5 µl - 5 µl sample volume
Lid	50	Lid 10	1 µl - 3 µl sample volume
Units	ng/µl	Lid 50	0.3 µl - 2 µl sample volume
Background Correction		Lid 100	0.3 µl - 2 µl sample volume
		Lid 250	0.3 µl - 2 µl sample volume

Note: A new Blank is recommended when changing the dilution lid.

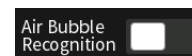
5. Select the Units in which the concentration should be calculated. Options are ng/µl (default), µg/µl and µg/ml. Molar units: pmol/µl is available only if a nucleic acid sequence is entered for nucleic acid factor calculation; if a custom molar extinction coefficient is specified, the unit of measurement will automatically be switched to nmol/l.

dsDNA	50	ng/µl
Volume	1 µl - 2 µl	µg/µl
Units	ng/µl	µg/ml

6. The background correction is enabled at 320 nm by default. Selection options are 320 nm, 340 nm or any wavelength in the range of 220 – 350 nm. The background correction can be disabled with the toggle switch.

dsDNA	50	Background Correction	☒
Volume	1 µl - 2 µl	Background Correction at 320 nm	
Units	ng/µl	Background Correction at 340 nm	
Background Correction		Background Correction at	320 nm

7. Air bubble recognition is disabled by default. Setting to on detects air bubbles, lint residues and bad sample conditions of the sample.



Note: Lint residues and bad sample conditions are detected even if the air bubble recognition is set to off.

8. For dye labelled samples tick the dye labels in the list, which should be used for result calculation.

Note: If the used dye is not available in the list, please go to preferences and add a custom dye to the dye list.

There is an option for dye correction which can be enabled/disabled with a toggle switch.

Note: Dye correction is only available for single dye selection.

dsDNA	50	Dye Correction ☐		
Volume	1 µl - 2 µl	☐ Alexa Fluor 350	☐ Alexa Fluor 647	☐ Oyster-500
Units	ng/µl	☐ Alexa Fluor 488	☐ Alexa Fluor 660	☐ Oyster-550
Background Correction		☐ Alexa Fluor 532	☐ Alexa Fluor 680	☐ Oyster-556
Air Bubble Recognition ☐		☐ Alexa Fluor 546	☐ Cy3	☐ Oyster-645
Dye Label		☐ Alexa Fluor 555	☐ Cy3.5	☐ Oyster-650
Manual Dilution		☐ Alexa Fluor 568	☐ Cy5	☐ Oyster-656
		☐ Alexa Fluor 594	☐ Cy5.5	☐ Texas Red

9. Option to set/calculate a dilution factor for manual diluted samples.

dsDNA	50	Dilution Factor	1
Volume	1 µl - 2 µl	Calculate Dilution Factor	☐
Units	ng/µl		
Background Correction			
Air Bubble Recognition ☐			
Dye Label			
Manual Dilution			

10. Apply the blank ddH₂O or buffer to the illuminated sample window on the pedestal and close the lid arm for the reference measurement and select blank to initiate the reading.

Note: The illumination of the sample window can be switched off in the preferences.



11. Use a lint-free laboratory wipe to clean both the sample window on pedestal and mirror in lid arm prior to applying the next sample.

Note: It could be helpful to apply the blank a second time and read it as a sample to ensure a proper blank.



12. Apply sample to the sample window on pedestal and press the sample button to initiate the measurement.

If auto sample is active the sample measurement is automatically started after closing the lid arm.



Note: Auto sample function is only available NanoVolume methods and new NanoPhotometer® versions it may not be available after updating an older firmware version.

MEASUREMENT PROTOCOL MULTI SAMPLE (N120)

1. Select the Nucleic Acids icon on the home screen.



2. To change between single and multi-sample application use the toggle switch Single Sample in the parameter area.



3. Parameter setting Nucleic Acid type:

To change the nucleic acid type, select either dsDNA, ssDNA, RNA, miRNA, miRNA Sequence, Oligo, Oligo Sequence or Custom (see Table 1 on page 66).

- The miRNA sequence and Oligo sequence options allow the sequence to be entered and the extinction coefficient will be automatically calculated. A minimum of 4 bases is required.
- The default setting for Custom is 50. The extinction coefficient ($1/\epsilon$) can be set in a range of 15 - 150. It is possible to enter a name for the custom nucleic acid factor for documentation.
- The default setting for Custom Molar Extinction Coefficient (Mol. Ext. Coeff.) is 10 and be set within a range of 0 - 15. It is possible to enter a name for the custom molar extinction coefficient for documentation.

dsDNA	50	dsDNA	50	ssDNA	37
Volume	1 µl - 2 µl	RNA	40	miRNA	33
Units	ng/µl	miRNA Seq.	Please enter miRNA sequence		
Background Correction		Oligo	33		
Air Bubble Recognition	☐	Oligo Seq. DNA	Please enter DNA Oligo sequence		
Dye Label		Oligo Seq. RNA	Please enter RNA Oligo sequence		
Manual Dilution		Custom	Please enter Name	Please enter Factor	
		Custom (Mol. Ext.Coeff.)	Please enter Name	Please enter Mol. Ext. Coeff.(nm ⁻¹ cm ⁻¹)	

4. Select the Units in which the concentration should be calculated.
Options are ng/µl (default), µg/µl and µg/ml. Molar units: pmol/µl is available only if a nucleic acid sequence is entered for nucleic acid factor calculation; if a custom molar extinction coefficient is specified, the unit of measurement will automatically be switched to nmol/l.

dsDNA	50	ng/μl
Units	ng/μl	μg/μl
Background Correction		μg/ml

5. The background correction is enabled at 320 nm by default. It is possible to disable the background correction with the toggle switch. Selection options are 320 nm, 340 nm. It is also possible to add any wavelength in the range of 220 – 350 nm

dsDNA	50	Background Correction ☒
Units	ng/μl	Background Correction at 320 nm
Background Correction		Background Correction at 340 nm
		Background Correction at 320 nm

6. Air bubble recognition is disabled by default. Set to on it detects air bubbles, lint residues and bad sample conditions of the sample.
Note: Lint residues and bad sample conditions are detected even if the air bubble recognition is set to off.

Air Bubble Recognition	☐
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7. For dye labelled samples tick the dye labels in the list, which should be used for result calculation.
Note: If the used dye is not available in the list, please go to preferences and add a custom dye to the dye list (see page 130 Dyes).
 There is an option for dye correction which can be enabled/disabled with a toggle switch.
Note: Dye correction is only available for single dye selection.

dsDNA	50	Dye Correction ☐
Units	ng/μl	
Background Correction		
Air Bubble Recognition	☐	
Dye Label		
Sample Loading		
Single Sample	☐	
	☐ Alexa Fluor 350 ☐ Alexa Fluor 647 ☐ Oyster-500 ☐ Texas Red ☐ Alexa Fluor 488 ☐ Alexa Fluor 660 ☐ Oyster-550 ☐ Alexa Fluor 532 ☐ Alexa Fluor 680 ☐ Oyster-556 ☐ Alexa Fluor 546 ☐ Cy3 ☐ Oyster-645 ☐ Alexa Fluor 555 ☐ Cy3.5 ☐ Oyster-650 ☐ Alexa Fluor 568 ☐ Cy5 ☐ Oyster-656 ☐ Alexa Fluor 594 ☐ Cy5.5 ☐ Test	

8. Select the sample loading orientation.

dsDNA	50	horizontally	(up to 12 samples e.g. A01 – A12)
Units	ng/μl	vertically	(up to 8 samples e.g. A01 – H01)
Background Correction			
Air Bubble Recognitor	☐		
Dye Label			
Sample Loading			

There are two options:

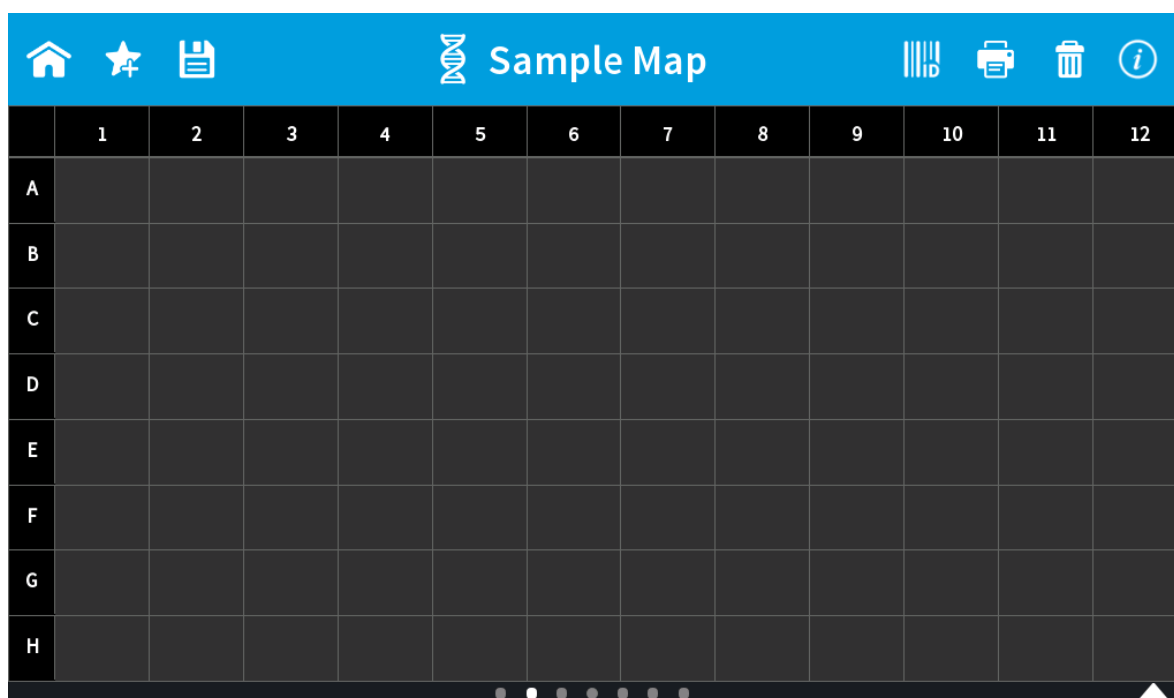
Horizontally: Measure up to 12 samples (A01 – A12 shown in sample map as row)

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												

Vertically: Measure up to 8 samples (A01 – H01 shown in sample map as column)

	1	2	3	4
A				
B				
C				
D				
E				
F				
G				
H				

- To confirm the parameter setting, swipe the screen to the left and the sample map screen is shown.

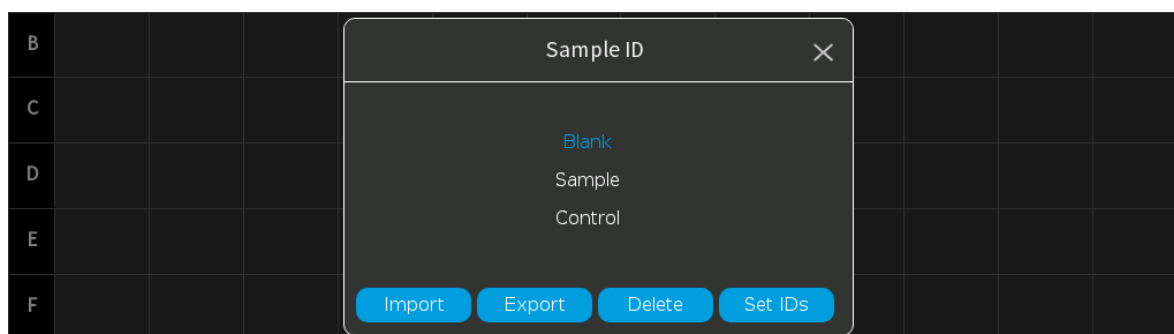


The sample map offers the possibility to select cells and define sample IDs for measurements. It is not necessary to make any selection. If cells are left empty without defined sample IDs for horizontal or vertical sample loading, all 12 positions for horizontal loading or all 8 positions for vertical loading are measured.

To measure a defined amount of samples select a cell pattern starting at A01.

Sample IDs can be defined by pressing the sample ID icon (|||||) in the navigation bar.

The opened pop up offers options to define, import, export or delete sample IDs.



Further information about cell selection and sample ID definition can be found in chapter 3 NanoPhotometer® Basics / Sample Map on page 36

10. Apply the blank ddH₂O or buffer to the illuminated sample windows on pedestal for the reference measurement and select blank to initiate the reading.



Note: The illumination of the sample window can be switched off in preferences.

Note: If the blank and sample buttons are not shown please either open the lid arm, push the ▲ on the right bottom corner or swipe from the

bottom edge in the screen.

11. Use a lint-free laboratory wipe to clean both the sample windows on pedestal and mirrors in lid arm prior to applying the next sample.

Note: It could be helpful to apply the blank a second time and read it as a sample to ensure a proper blank.



12. Apply samples to the illuminated sample windows on pedestal and press the sample button to initiate the measurement.

If auto sample is active the sample measurement is automatically started after closing the lid arm.

Note: Auto sample function is only available NanoVolume methods and new NanoPhotometer® versions it may not be available after updating an older firmware version.



CALCULATIONS

▪ Nucleic Acid Concentration

To determine the concentration of nucleic acids in solution, the absorbance is measured at a wavelength of 260 nm. The function describing the relationship between concentration and absorbance is a modification of the Beer-Lambert law equation. The concentration of the nucleic acid samples can be calculated with or without background correction depending on enabled/disabled background correction parameter.

Without background correction:

$$C = A_{260} * \text{Factor}_{\text{nuc}} * D$$

With background correction:

$$C = (A_{260} - A_{\text{BKG}}) * \text{Factor}_{\text{nuc}} * D$$

C	Concentration in ng/μl
A ₂₆₀	Absorbance at 260 nm (10 mm path)
A _{BKG}	Absorbance at selected background wavelength (10 mm path)
D	Manual dilution factor
Factor _{nuc}	Nucleic acid factor in ng*cm/μl

Table 1. Nucleic acids factors (Factor_{nuc})

Type	Factor _{nuc}
dsDNA	50 ng*cm/μl
ssDNA	37 ng*cm/μl
RNA	40 ng*cm/μl
miRNA	33 ng*cm/μl
Oligo	33 ng*cm/μl
miRNA Seq.	calculated via extinction coefficient of constituent nucleotides entered

Oligo Seq.	calculated via extinction coefficient of constituent nucleotides entered
Custom	Option to enter any factor between 15 and 150 ng*cm/μl
Custom (Mol. Ext. Coeff.)	Option to enter any molar extinction coefficient between 0 and 15 1/nM*cm

▪ Dye-labeled Nucleic Acid Concentration

For dye-labeled nucleic acids, the concentration of the nucleic acid is calculated using a modified form of the Beer-Lambert equation. For these calculations, the instrument considers the absorption maximum of the dye and a certain dye-specific correction factor at 260 nm (see Table 2 on page 68). The concentration of a dye-labeled nucleic acid is calculated with or without background/dye correction as follows:

With background and with dye correction:

$$C = \left[(A_{260} - A_{BKG}) - (cf_{dye} * (A_{max, dye} - A_{BKG})) \right] * Factor_{nuc} * D$$

With background and without dye correction:

$$C = (A_{260} - A_{BKG}) * Factor_{nuc} * D$$

Without background and with dye correction:

$$C = [A_{260} - (cf_{dye} * A_{max, dye})] * Factor_{nuc} * D$$

Without background and without dye correction:

$$C = A_{260} * Factor_{nuc} * D$$

C	Concentration in ng/μl
A ₂₆₀	Absorbance at 260 nm (10 mm path)
A _{BKG}	Absorbance at selected background wavelength (10 mm path)
A _{max, dye}	Absorbance value at the absorbance maximum of the dye (10 mm path)
Factor _{nuc}	Nucleic acid factor in ng*cm/μl
cf _{dye}	Dye-dependent correction factor at 260 nm
D	Manual dilution factor

Note: Dye correction is only available for single dye selection.

▪ Dye Concentration

For dye-labeled nucleic acids, the concentration of the dye is calculated using a modified form of the Beer-Lambert equation. For these calculations, the instrument considers the absorption maximum of the dye, the dye-specific extinction coefficient (see Table 2 on page 68). The dye concentration is calculated with or without background correction as follows:

With background correction:

$$C = \frac{(A_{max, dye} - A_{BKG}) * D}{m\epsilon_{dye} * 10^{-6}}$$

Without background correction:

$$C = \frac{A_{\max, \text{dye}} * D}{m\epsilon_{\text{dye}} * 10^{-6}}$$

C	Concentration in pmol/μl
$A_{\max, \text{dye}}$	Absorbance value at the absorbance maximum of the dye (10 mm path)
A_{BKG}	Absorbance at 320 nm (10 mm path)
$m\epsilon_{\text{dye}}$	Molar extinction coefficient of dye in $\text{M}^{-1} * \text{cm}^{-1}$
D	Manual dilution factor

▪ **Frequency of Incorporation (FOI)**

FOI is the degree of labeling based on dye incorporation in a labeled nucleic acid sample. It is generally expressed as the number of dye molecules incorporated per 1000 nucleotides. FOI can be calculated with or without background/dye correction as follows:

With background correction with dye correction:

$$\text{FOI} = \frac{324.5 * (A_{\max, \text{dye}} - A_{\text{BKG}})}{m\epsilon_{\text{dye}} * 10^{-6} * (A_{260} - A_{\text{BKG}} - c_{\text{dye}} * (A_{\max, \text{dye}} - A_{\text{BKG}})) * \text{Factor}_{\text{nuc}}}$$

With background correction without dye correction:

$$\text{FOI} = \frac{324.5 * (A_{\max, \text{dye}} - A_{\text{BKG}})}{m\epsilon_{\text{dye}} * 10^{-6} * (A_{260} - A_{\text{BKG}}) * \text{Factor}_{\text{nuc}}}$$

Without background correction and with dye correction:

$$\text{FOI} = \frac{324.5 * A_{\max, \text{dye}}}{m\epsilon_{\text{dye}} * 10^{-6} * (A_{260} - c_{\text{dye}} * A_{\max, \text{dye}}) * \text{Factor}_{\text{nuc}}}$$

Without background correction and without dye correction:

$$\text{FOI} = \frac{324.5 * A_{\max, \text{dye}}}{m\epsilon_{\text{dye}} * 10^{-6} * A_{260} * \text{Factor}_{\text{nuc}}}$$

FOI	Frequency of Incorporation (dye per 1,000 bases)
A_{260}	Absorbance at 260 nm (10 mm path)
A_{BKG}	Absorbance at selected background wavelength (10 mm path)
$A_{\max, \text{dye}}$	Absorbance value at the absorbance maximum of the dye (10 mm path)
$m\epsilon_{\text{dye}}$	molar extinction coefficient of dye in $\text{M}^{-1} * \text{cm}^{-1}$
$\text{Factor}_{\text{nuc}}$	Nucleic acid factor in ng*cm/μl
c_{dye}	Dye-dependent correction factor at 260 nm

Table 2. Dye Types, Absorbance Max, Extinction coefficient, and dye-dependent correction factors

NanoPhotometer® Model	Dye Type	Absorbance maximum of Dye (nm)	molar ext. coeff. of dye $m\epsilon_{\text{dye}}$ in $\text{M}^{-1}\cdot\text{cm}^{-1}$	Dye-dependent correction factor at 260 nm cf_{dye}
NP80/N60/N50/C40	Alexa Fluor 350	346	19,000	0.25
NP80/N60/N50/C40	Alexa Fluor 488	495	71,000	0.30
NP80/N60/N50/C40	Alexa Fluor 532	532	81,000	0.24
NP80/N60/N50/C40	Alexa Fluor 546	554	112,000	0.21
NP80/N60/N50/C40	Alexa Fluor 555	555	150,000	0.08
NP80/N60/N50/C40	Alexa Fluor 568	578	91,300	0.45
NP80/N60/N50/C40	Alexa Fluor 594	590	90,000	0.43
NP80/N60/C40	Alexa Fluor 647	650	239,000	0.00
NP80/N60/C40	Alexa Fluor 660	663	132,000	0.00
NP80/N60/C40	Alexa Fluor 680	679	184,000	0.00
NP80/N60/N50/C40	Cy3	550	150,000	0.08
NP80/N60/N50/C40	Cy3.5	581	150,000	0.08
NP80/N60/C40	Cy5	649	250,000	0.05
NP80/N60/C40	Cy5.5	675	250,000	0.05
NP80/N60/N50/C40	Oyster-500	503	78,000	0.29
NP80/N60/N50/C40	Oyster-550	553	150,000	0.05
NP80/N60/N50/C40	Oyster-556	560	155,000	0.03
NP80/N60/C40	Oyster-645	649	220,000	0.05
NP80/N60/C40	Oyster-650	653	200,000	0.04
NP80/N60/C40	Oyster-656	660	200,000	0.04
NP80/N60/N50/C40	Texas Red	603	112,000	0.23

▪ Ratios

Reactions utilizing nucleic acids often require minimum purity standards. Common contaminants of nucleic acid samples include: proteins, organic compounds, and other. Based on the common contaminants of nucleic acid samples, the 260/280 and 260/230 ratios are calculated for nucleic acids to give an indication of the purity of the samples. Pure DNA and RNA preparations have expected 260/280 ratios of ≥ 1.8 and ≥ 2.0 respectively. An elevated absorbance at 230 nm can indicate the presence of impurities as well; 230 nm is near the absorbance maximum of peptide bonds and also indicates buffer contamination since TRIS, EDTA and other buffer salts absorb at 230nm. When measuring RNA samples, the 260/230 ratio should be > 2.0 ; a ratio lower than this is generally indicative of contamination with guanidinium thiocyanate, a reagent commonly used in RNA purification and which absorbs over the 230-260 nm range. If a ratio is detected out of the acceptable range an alert icon  is shown in the results/table area. A push on the alert icon shows additional information. The ranges for acceptable ratio values can be defined in preferences. The ratios are calculated with or without background correction according to if the background correction is activated during the measurements or not as follows:

Without background correction:

$$260/280 \text{ ratio} = \frac{A_{260}}{A_{280}}$$

$$260/230 \text{ ratio} = \frac{A_{260}}{A_{230}}$$

With background correction:

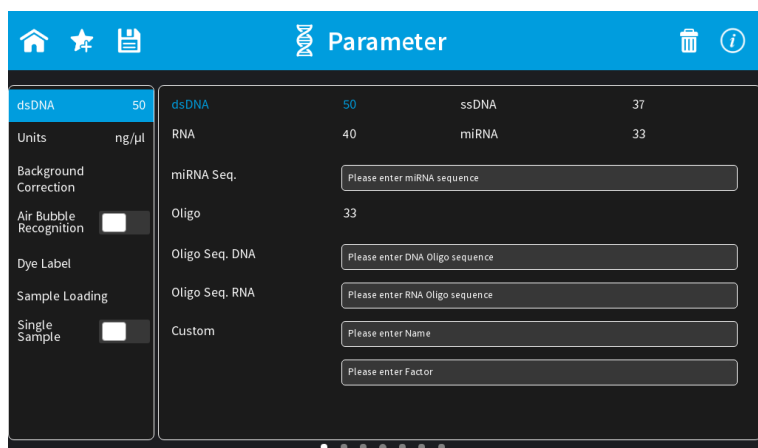
$$260/280 \text{ ratio} = \frac{A_{260} - A_{BKG}}{A_{280} - A_{BKG}}$$

$$260/230 \text{ ratio} = \frac{A_{260} - A_{BKG}}{A_{230} - A_{BKG}}$$

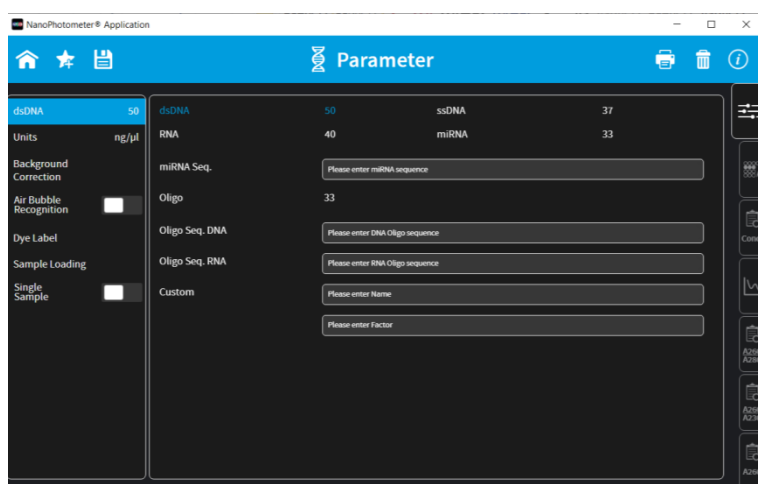
MULTI SAMPLE METHODS

Change Screens

Opening a multi sample method (Nucleic Acid, Protein UV, Protein Assay or Wavescan) shows the parameter screen first. To confirm the parameter swipe the screen to the left and the sample map is shown. It is possible to change between the parameter, sample map and result screens by swiping left or right at any time. At the bottom of the screen is a screen carousel that shows the active and available screens.



For computer versions without touchscreen a vertical tab bar is shown to change between the different screens (≡ parameter, ≡ sample map, ≡ results and ≡ graph; for Protein Assay additionally ≡ standard map and ≡ standard curve)

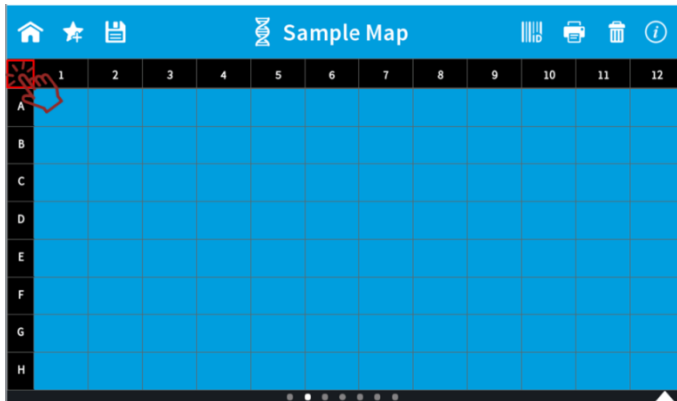


Sample Map / Definition of Sample IDs

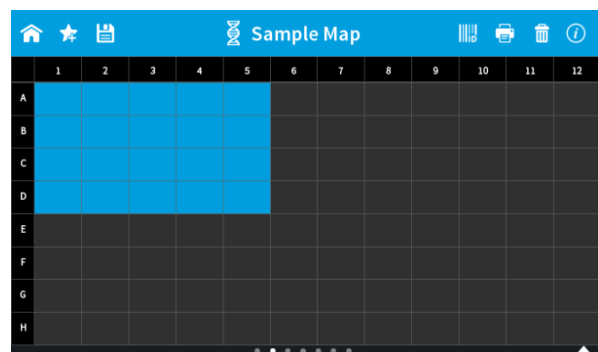
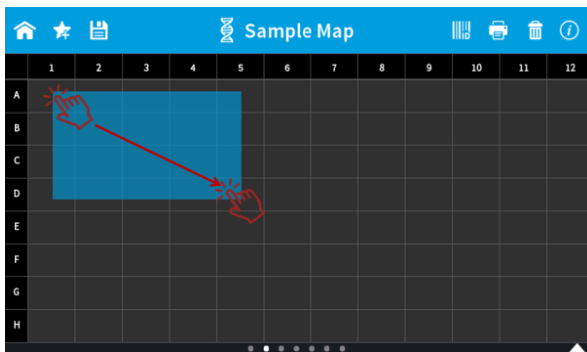
The sample map screen shows the sample IDs as defined or names it automatically if nothing is defined. For measurements it is not necessary to select any cell or define sample IDs. If no cell is selected/Sample ID defined, all 12 (horizontally) or 8 (vertically) samples are measured. Sample IDs are named automatically in this case.

Selection Options

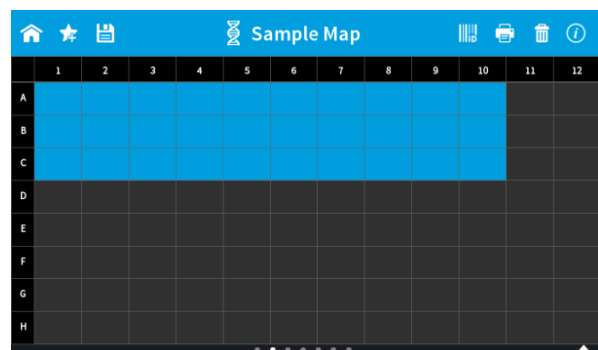
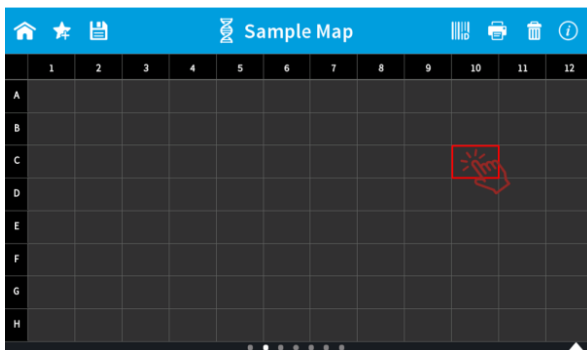
Selection of cells is only possible in the sample map screen. Cell A01 must always be selected. A push in the top left corner selects/deselects all 96 cells at once:



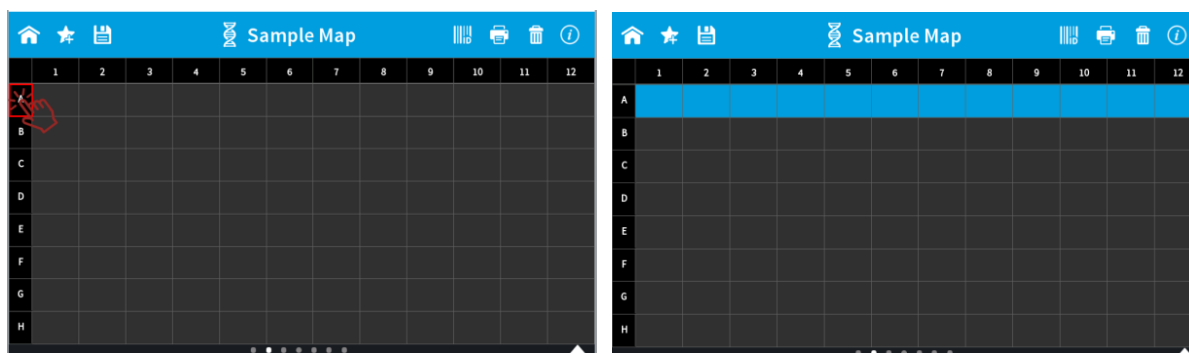
Manual selection is possible by selecting any area starting in cell A01.



or by tapping any cell:



A push to the row/column header selects/deselects the whole row/column:

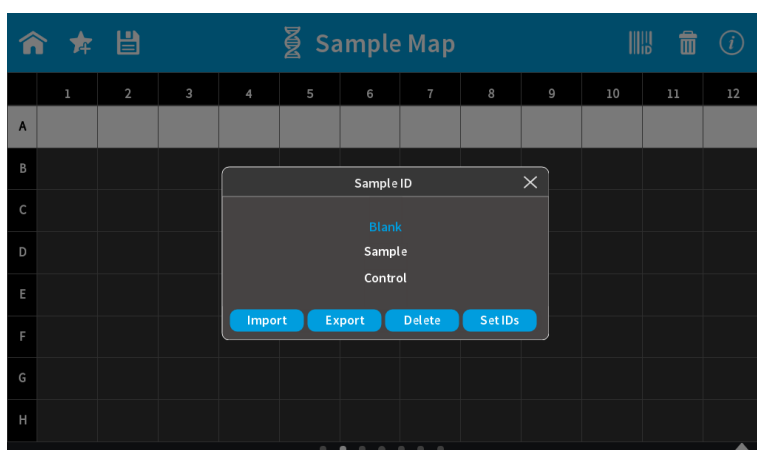


Note: As a blank is necessary for each measurement it is not possible to select more cells as in the previous row/column.

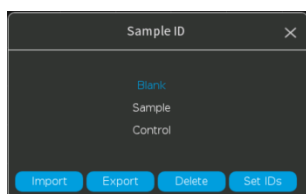
Sample ID Definition

There are multiple options to define Sample IDs. If no Sample IDs are defined the Sample IDs are named automatically.

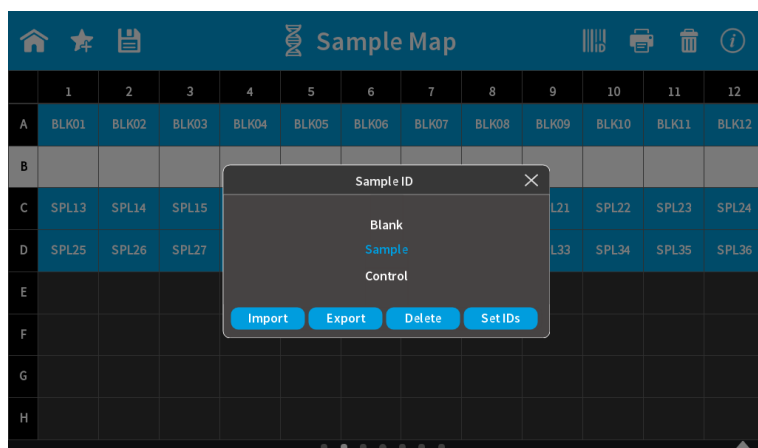
To define sample IDs open the Sample ID pop up (⋮) on the Sample Map screen. The Sample ID pop up offers options to set sample IDs, delete sample IDs, import or export sample IDs from a csv file.



To set sample IDs, select Blank (BLK), Sample (SPL) or Control (CRL) and push on “Set IDs” button. Sample IDs are numbered automatically.



To delete sample IDs, close the pop up, select the row/column for which the sample IDs should be deleted and reopen the sample ID pop up and push the Delete button.



Individual Sample IDs can be changed by double-clicking on the Sample ID.

To change several Sample IDs simultaneously, open the table view after a measurement and select Edit.

Sample IDs can be imported with the Import function. For the import of Sample IDs, the sample IDs must first be defined in the following format and saved as a csv file.

- for horizontal Sample Map:

	A	B	C	D	E	F	G	H	I	J	K
1	Position	Content	Sample ID		Comments						
2	A01	B	BLK01								
3	A02	B	BLK02		Content field needs to be defined with B, C or S						
4	A03	B	BLK03		B = Blank measurement						
5	A04	B	BLK04		C = Control measurement						
6	A05	B	BLK05		S = Sample measurement						
7	A06	B	BLK06		It is not possible to mix Blank and Control/Sample measurements in one row.						
8	A07	B	BLK07		Control and Sample contents can be combined in one row.						
9	A08	B	BLK08								
10	A09	B	BLK09		Blank and Control have pre-defined sample IDs						
11	A10	B	BLK10		B = BLK01, BLK02, etc.						
12	A11	B	BLK11		C = CTRL01, CTRL02, etc.						
13	A12	B	BLK12		If anything else than the pre-defined sample IDs is entered for Blank and Control, the sample IDs for samples can be user-defined, if no sample IDs are defined for content (S) then S = SPL01, SPL02, etc.						
14	B01	C	CTRL01								
15	B02	C	CTRL02								
16	B03	C	CTRL03								
17	B04	C	CTRL04		It is not necessary to set samples 1 - 12, any number > 1 is possible						
18	B05	C	CTRL05								
19	B06	C	CTRL06								

Note: Draft files are available for download on the Implen homepage (www.implen.de/downloads).

- for vertical Sample Map:

	A	B	C	D	E	F	G	H	I	J	K
1	Position	Content	Sample ID		Comments						
2	A01	B	BLK01								
3	B01	B	BLK02		Content field needs to be defined with B, C or S						
4	C01	B	BLK03		B = Blank measurement						
5	D01	B	BLK04		C = Control measurement						
6	E01	B	BLK05		S = Sample measurement						
7	F01	B	BLK06		It is not possible to mix Blank and Control/Sample measurements in one column						
8	G01	B	BLK07		Control and Sample contents can be combined in one column						
9	H01	B	BLK08								
10	A02	C	CTRL01		Blank and Control have pre-defined sample IDs						
11	B02	C	CTRL02		B = BLK01, BLK02, etc.						
12	C02	C	CTRL03		C = CTRL01, CTRL02, etc.						
13	D02	C	CTRL04		If anything else than the pre-defined sample IDs is entered for Blank and Control, the sample IDs for samples can be user-defined, if no sample IDs are defined for content (S) then S = SPL01, SPL02, etc.						
14	E02	C	CTRL05								
15	F02	C	CTRL06								
16	G02	C	CTRL07								
17	H02	C	CTRL08		It is not necessary to set samples 1 - 08, any number > 1 is possible						
18	A03	S									
19	B03	S									
20	C03	S									

Note: Draft files are available for download on the Implen homepage (www.implen.de/downloads).

Note: It is not possible to edit imported Sample IDs.

To import sample IDs open the sample ID pop up and push the import button. A file directory is shown. Select the .csv file and push on import button. Sample IDs will be imported.

To export sample IDs, press the export button in the sample ID pop up. Enter a file name, select a folder and confirm with export.

▪ Result Screens

There are multiple result screens available depending on parameter setting and method.

Nucleic Acids:

- Concentration
- Graph
- Ratio A260/A280
- Ratio A260/A230
- A260

Protein UV:

- Concentration
- Graph
- Ratio A260/A280
- A280

If a dye is added three additional screens are shown:

- Dye Concentration
- Dye Absorbance
- FOI / DOL

Protein Assay:

- Standard Map
- Standard Curve
- Concentration
- Graph
- Absorbance

Wavelength:

- Graph

Wavescan:

- Graph

Defined sample IDs are shown on all result screens as long as the sample is not measured. Once the sample is measured the result is shown. A white frame marks the sample IDs which will be measured next.